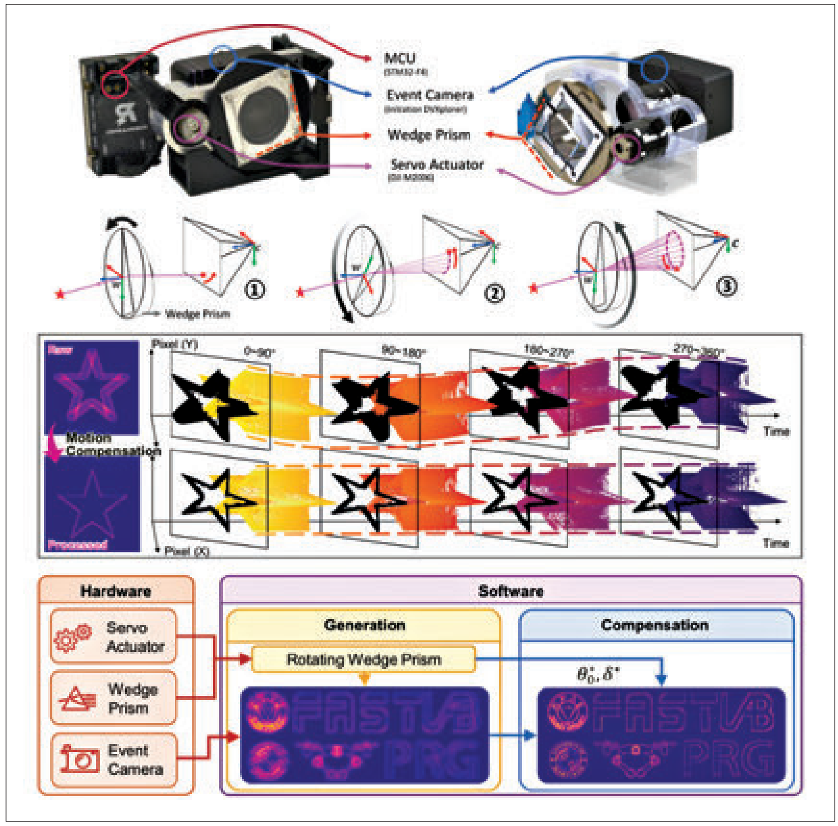
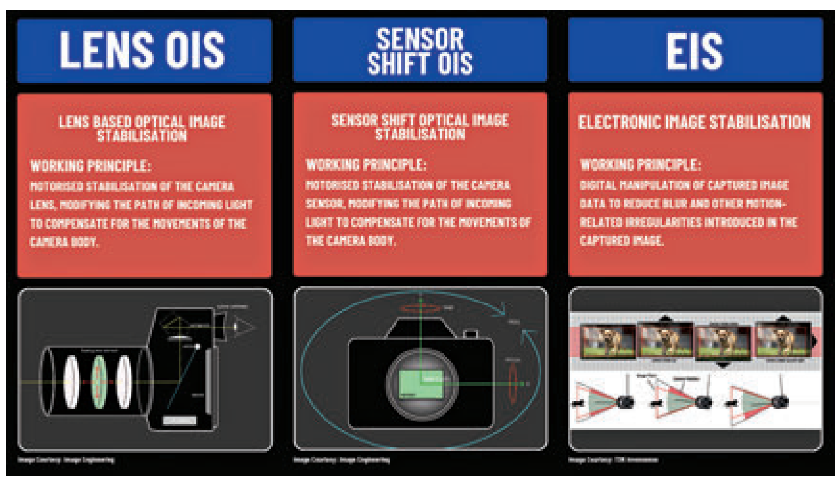
eras which feature individually operating pixels, which capture and respond to incoming light data and report brightness changes on the fly. These cameras are also called – neuromorphic cameras, silicon retina or dynamic vision sensors.

Once they had figured out the mechanics of it and had the hardware in place, the next challenge lay in the form of developing software that could process this data. And they were able to do that successfully as well. The primary aim of the software, as published in an article covering this innovation by UMD, was "to compensate for the prism's movement within the AMI-EV to consolidate stable images from the shifting lights".

After this, the researchers were able to successfully capture and display movement accurately in a variety of contexts. This included the detection of human pulse and shape identification of rapidly moving subjects. The most astonishing part lay in the number of frames that AMI-EV was able to capture. As stated in the research article, tens of thousands of frames per second, which is significantly higher than that of the cameras that are commercially available today, capable of capturing anywhere between 30-1000 frames per second.





Where do the roads lead to?

Well, if you look at it, AMI-EV can have applications beyond what we would think of after learning about this innovation. The researchers have high hopes that this new imaging technology will provide breakthroughs in applications that are not only limited to smartphones and robotics but also beyond. Commenting on this, Cornelia Fermüller, a senior author of the paper, said, "With their unique features, event sensors and AMI-EV are poised to take center stage in the realm of smart wearables." Fermüller added, "They have distinct advantages over classical cameras—such as superior performance in extreme lighting conditions, low latency and low power consumption. These features are ideal for virtual reality applications, for example, where a seamless experience and the rapid computations of head and body movements are necessary."

It was further stated that AMI-EV could find its way into self-driving automobiles, enabling much more efficient and faster recognition of objects on the road. In terms of smartphones, the claimed power efficiency of the mechanism, coupled with its capabilities to capture clearer and crisper videos and photos and improved low-light performance, could make AMV-EV an avenue that manufacturers would definitely consider globally. Additionally, this technology could enhance security imaging and further the way in which astronomers capture images in space.

The possibilities are vast, and there is potential in AMI-EV. Now, it is up to researchers across the world to probe further into this and experiment with its different applications to ensure it improves the way we capture visual information. d